

Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper is the addendum (-AddA) to the parent **Paper-14: A conditional BRST gauge-fixing and Faddeev-Popov closure note**, written in response to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Canonical TECT positioning (binding 2026-05-26): *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* TOE-level claim SUSPENDED until non-standard rescue independently derived.

I. BACKGROUND AND PARENT-PAPER RECAP

Paper-14 presented the Math160 BRST framework as a conditional gate closure for GAP-2 (BRST quantization consistency). Two technical components were combined: (i) the FP determinant construction on the $SO(10)$ bundle over TECT's BCC base manifold; (ii) the Math226 demonstration that the standard topological obstruction to BRST consistency — non-trivial π_1 of the BRST configuration space — is absent because $\pi_1(M_{\text{BCC}}) = \{e\}$ (all loops contractible).

II. MATHEMATICAL CONTENT SURVIVAL

Component (i) — the FP determinant construction — is conditional on $SO(10)$ gauge structure being realised. The 2026-05-26 reclassification refutes the standard-framework dynamical realisation but preserves the structural lemma: “GIVEN $SO(10)$ realised, the FP determinant is well-defined.” T6 PROVED CONDITIONAL on H1 ($SO(10)$ realisation by any mechanism). **Component (ii)** — the Berry-phase signature triviality — is a TECT-specific topological statement about M_{BCC} , independent of Pillar 6. T7 PROVED (within standard π_1 definition; higher-form topology rescue Math164 §3 separately tracked).

III. RECLASSIFICATION IMPLICATIONS

Post-reclassification, GAP-2 status: **T4 STRONG EVIDENCE**, conditional on Pillar 6 non-standard-rescue derivation. The previous status (T6 PROVED CONDITIONAL on Pillar 4 sub-task 1) is downgraded because the conditional-on chain now includes the queued non-standard Pillar 6 rescue. The downgrade is structural (not new evidence), consistent with §6.3.5(b) ε -type-inequality

classification: the result is no longer a constant-bound theorem; it is an estimate awaiting the missing rescue.

IV. FORWARD PATH: NON-STANDARD RESCUES

If Math411-AddB-AddA fermion-condensate aux channels or Math411-AddB-AddB holographic emergence are independently derived and produce $SO(10)$, the Math160 FP determinant becomes well-defined as a T6 PROVED CONDITIONAL theorem on the rescue's hypothesis set. Until then, GAP-2 remains T4 STRONG EVIDENCE.

V. CROSS-REFERENCES

- Math160 (canonical BRST FP determinant note)
- Math226 (Berry-phase signature triviality)
- Math164 §3 (higher-form Berry topology, queued rescue)
- Paper-14 (parent paper)
- Math411-AddB §10 (2026-05-26 reclassification)